MIL-D-52328B
1 June 1973
SUPERSEDING
MIL-D-52328A
29 July 1964

MILITARY SPECIFICATION

DEPRESSOR ASSEMBLY, HYDRAULIC, BALK: FOR BRIDGE, FLOATING: ALUMINUM, HIGHWAY; DECK BALK SUPERSTRUCTURE

This specification is **approved for use by all Departments and Agencies of the Department** of Defense.

1. SCOPE

1.1 This specification covers a hydraulic balk depressor assembly used in the erection of fixed and floating bridges having aluminum deck balk superstructures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

Military

MIL-T-704	- Treatment and Painting of Materiel.

FSC 5420

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

DRAWINGS

ME

TA13219E3811	- Depressor Assembly, Hydraulic, Balk.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

A5.1	- Mild Steel Covered Arc-Welding Electrodes.
A5.10	- Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.
A5.18	- Mild Steel Electrodes for Gas Metal-Arc Welding.
B3.0	- Standard Qualification Procedure.
D1.1	- Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Description. The depressor assembly shall be as shown on Top Assembly TA13219E3811 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish a depressor assembly for inspection within the time frame specified (see 6.2) to prove prior to starting production, that his production methods will produce depressor assemblies that comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Welding electrodes.

3.4.1 For steel. Welding electrodes for steel shall conform to AWS A5.1 or A5.18, classification E60 or E70 series, size as required.

3.4.2 For aluminum. Welding electrodes for aluminum shall conform to AWS A5.10, classification ER4043, size as required.

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3.5 Identification marking. The depressor assembly shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. The depressor assembly shall be cleaned, treated, and painted in accordance with MIL-T-704, type B. The contact surfaces of all dissimilar metals shall be painted prior to assembly.

3.7 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each depressor. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.8 Workmanship.

3.8.1 Steel fabrication. Steel may be cut by shearing, sawing, or flame cutting. Burned surfaces of flame-cut material shall be ground or machined to remove cooling checks. All bends shall be made by use of metal dies or fixtures to insure uniformity in size and shape. All burrs shall be removed.

3.8.2 Aluminum fabrication. Aluminum used in the fabrication of the depressor assembly shall be free from kinks and sharp bends. Straightening of material shall be done by methods that will not cause injury to the metal. Surfaces and edges of all finished members shall be smooth and free from kinks, bends, winds, and excessive scratches. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Sharp instruments shall not be used for marking bend lines. Aluminum shall not be cut by methods using heat. Aluminum less than 3/8 inch thick may be cut by shearing; aluminum 3/8 inch thick and over shall be cut by routing or by high-speed saws. Shearing and sawing shall be done neatly and accurately.

3.8.3 Rivet holes. Holes for countersunk rivets shall be of such depth and angle that the rivet heads shall fill the hole without projecting or recessing in excess of 1/32 inch.

3.8.4 Welders and welding.

3.8.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

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Standard Qualification Procedure of the American Welding Society (B3.0). Welding Qualification of the ASME Boiler and Pressure Vessel Code (Section IX).

3.8.4.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication, "D1.1 - Structural Welding Code." All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wire brushed clean.

3.8.4.2.1 Steel welding. Where embrittlement of the joined metal may result, preheating or annealing or both shall be employed.

3.8.4.2.2 Aluminum welding. Aluminum members to be welded may be preheated for welding to a temperature of not more than 400° F. Aluminum welds may be ground, filed, wire brushed, or chipped but shall not be hammered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Acceptability criteria. Depressor assemblies which conform to all requirements in sections 3 and 5 of this specification and pass all applicable examinations in section 4 of this specification will be considered acceptable by the Government.

4.1.2 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction depressor assembly shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The depressor assembly shall be examined as specified herein, for the following defects:

Major

101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.
102. Material not as specified.
103. Components missing or not as specified.
104. Workmanship not as specified.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.
106. Riveting not as specified.

Minor

201. Dimensions other than those identified by SQAP data not as specified.
202. Identification marking incorrect, missing, or illegible.
203. Treatment and painting not as specified.
204. Stenciling not as specified.

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4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Hydraulic fluid reservoir not filled, cylinder retracted, and valve closed as specified.
108.	Boxes not as specified for level A or B.
109.	Depressor assemblies packed together in quantities other than specified for level A or B.
110.	Blocking and bracing not adequate to prevent movement or damage for level A or B.
111.	Depressors or jacks procured separately not packed as specified for level A or B.
112.	Strapping not zinc coated for level A.
113.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 Preservation (see 6.6). The hydraulic fluid reservoir of each jack shall be filled to the operating level with the fluid recommended for operation. The cylinder shall be retracted to minimum length and the valve closed. Preservation and packaging of other components of the assembly shall not be required.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four complete depressor assemblies shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, grade B or PPP-B-621, class 2, style optional. The contents shall be blocked and braced within the box to prevent movement or damage. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated.

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5.2.1.1 Depressors or jacks (when procured separately). When depressors or jacks are procured separately from the depressor assembly, they shall be packed as specified in 5.2.1, except the quantities packed together shall be limited only to the weight limitation of the box specification.

5.2.2 Level B. The depressor assemblies or individual depressors or jacks when procured separately shall be packed as specified in 5.2.1 for level A except boxes shall be domestic type or class as applicable and strapping need not be zinc coated.

5.2.3 Level C. The depressor assemblies or individual depressors or jacks when procured separately shall be packed in a manner to assure carrier acceptance and safe delivery to destination at lowest rating in wooden or fiberboard containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking (see 6.6). Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hydraulic balk depressor assembly is intended for use in the erection of fixed and floating bridges having aluminum deck balk superstructure to depress the balk to facilitate placement of the pins.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model (see 3.2).
(c) Level of packing required (see 5.2).

6.3 Preproduction model. Any changes or deviations of production depressor assemblies from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish depressor assemblies conforming to this specification.

6.4 Supplementary Quality Assurance Provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, and corrected action taken when applicable."

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6.5 Definitions.

6.5.1 Supplementary Quality Assurance Provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.6 Levels of preservation have not been included as the requirements specified are the minimum acceptable for any level. For the purposes of preservation/packing level marking, the boxes shall be marked A/A, A/B, or A/C as applicable.

Custodians:

Army - ME

User activity:

Navy - MC

Preparing activity:

Army - ME

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-D-52328B Depressor Assembly, Hydraulic, Balk: For Bridge, Floating: Aluminum, Highway; Deck Balk Superstructure

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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